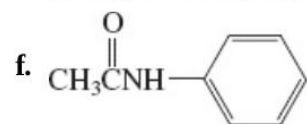
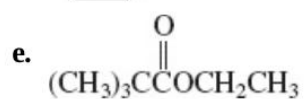
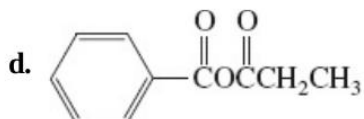
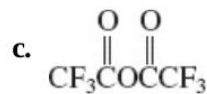
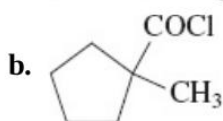
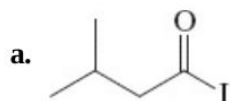


30. Name (IUPAC system) or draw the structure of each of the following molecules.



g. Propyl butanoate

h. Butyl propanoate

i. 2-Chloroethyl benzoate

j. *N,N*-Dimethylbenzamide

k. 2-Methylhexanenitrile

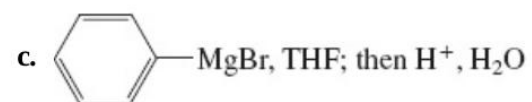
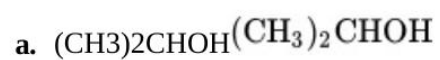
l. Cyclopentanecarbonitrile

32. **(a)** Use resonance forms to explain in detail the relative order of acidity of carboxylic acid derivatives, as presented in [Section 20-1](#). **(b)** Do the same, but use an argument based on inductive effects.

34. Give the product(s) of each of the following reactions.

35. Formulate a mechanism for the reaction of acetyl chloride with 1-propanol shown on [p. 975](#).

36. Give the product(s) of the reactions of acetic anhydride with each of the following reagents. Assume in all cases that the reagent is present in excess.



- 38.** Formulate a mechanism for the reaction of butanedioic (succinic) anhydride with methanol shown on [p. 978](#).

39. Give the products of reaction of methyl pentanoate with each of the following reagents under the conditions shown.

- a. $\text{NaOH}, \text{H}_2\text{O}, \text{H}_2\text{O}, \text{heat}; \text{then } \text{H}^+, \text{H}_2\text{O}$
- b. $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2\text{OH} (\text{CH}_3)_2\text{CHCH}_2\text{CH}_2\text{OH} (\text{excess}), \text{H}^+$
- c. $(\text{CH}_3\text{CH}_2)_2\text{NH}, (\text{CH}_3\text{CH}_2)_2\text{NH}, \text{heat}$
- d. $\text{CH}_3\text{MgI}, \text{CH}_3\text{MgI} (\text{excess}), (\text{CH}_3\text{CH}_2)_2\text{O}; (\text{CH}_3\text{CH}_2)_2\text{O}; \text{then } \text{H}^+, \text{H}_2\text{O}$
- e. $\text{LiAlH}_4, (\text{CH}_3\text{CH}_2)_2\text{O}; \text{LiAlH}_4, (\text{CH}_3\text{CH}_2)_2\text{O}; \text{then } \text{H}^+, \text{H}_2\text{O}$
- f. $[(\text{CH}_3)_2\text{CHCH}_2]_2\text{AlH}, [(\text{CH}_3)_2\text{CHCH}_2]_2\text{AlH}, \text{methylbenzene (toluene), low temperature}; \text{then } \text{H}^+, \text{H}_2\text{O}$

40. Give the products of reaction of γ -valerolactone (5-methyloxa-2-cyclopentanone, [Section 20-4](#)) with each of the reagents in [Problem 39](#).

42. Formulate a mechanism for the acid-catalyzed transesterification of ethyl 2-methylpropanoate (ethyl isobutyrate) into the corresponding methyl ester. Your mechanism should clearly illustrate the catalytic role of the proton.

- 43.** Formulate a mechanism for the reaction of methyl 9-octadecenoate with 1-dodecanamine shown on [p. 984](#).

44. Reaction review. Suggest reagents to convert each of the following starting materials into the indicated product: **(a)** acetyl chloride into acetic hexanoic anhydride; **(b)** methyl hexanoate into *N*-methylhexanamide; **(c)** hexanoyl chloride into hexanal; **(d)** hexanenitrile into hexanoic acid; **(e)** hexanamide into hexanamine; **(f)** hexanamide into pentanamine; **(g)** ethyl hexanoate into 3-ethyl-3-octanol; **(h)** hexanenitrile into 1-phenyl-1-hexanone [$\text{C}_6\text{H}_5\text{CO}(\text{CH}_2)_4\text{CH}_3$].

45. Give the product of each of the following reactions.

